

Paper G-03: AI Emergence: A Constraint-Driven Flux Dynamics (CDFD) Perspective

Steve Bico Mujjabi, MD

Independent Researcher

ORCID: 0009-0001-0556-5516

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Abstract

This paper asks whether Constraint-Driven Flux Dynamics (CDFD) and Adaptive Flux Limitation (AFL) can give the named system a sharper audit language. The working variables are driving flux Φ , constraint C , surface responsiveness S , and structural memory M_s . The useful question is plain: what can be measured, what would count as overload, and what result would make the mapping fail?

1 Series Context

Part I sets the flow–constraint language used here [1]. Part II develops the Life Number and the origins-of-life use of the same notation [2]. Part III applies AFL to biology and medicine [3]. Part IV is the cross-domain stress test: it asks where the notation clarifies measurement, and where it becomes only a metaphor.

2 Introduction

Notation. In Part IV, C names the active constraint or capacity burden in the system being discussed. Φ is the driving flow, S is the surface response, and M_s is retained structural memory. Λ appears only where the Part II Life Number is being referenced [2]. Other Greek symbols are local coefficients whose meaning is set by the equation in which they appear.

2.1 The Mujjabi Universal Capacity Law

Building directly upon the Fundamental Physics (Part I) [1], the **Mujjabi Universal Capacity Law** proposes that across all scales—from black hole event horizons to global financial markets and power grids—driving high flux (Φ) against a rigid topological constraint (C) can drive system saturation. When Φ/C approaches critical thresholds, the system does not scale linearly; it undergoes a topological phase transition.

2.2 The Mujjabi Universal Memory Law

The **Mujjabi Universal Memory Law** frames persistent systemic collapse. When a complex network fails to clear its structural memory (M_s) and its relaxation parameter decays ($\tau_{relax} \rightarrow \infty$), temporary stressors become permanent rigidities. This is the proposed CDFD mechanism behind ecological death, economic depressions, and infrastructural gridlock.

Artificial neural networks simulate learning by iteratively adjusting millions or billions of parameters to minimize a loss function against a training dataset. The current paradigm of AI development relies on massive brute-force scaling: pumping enormous volumes of data through immense clusters of GPUs.

While this has yielded emergent cognitive behaviors, the trajectory is fundamentally thermodynamic. The network is essentially an active engine consuming information and energy to produce structured mathematical memory. In CDFD terms, a deep learning model is an adaptive topological surface attempting to balance immense throughput against strict physical and architectural bottlenecks.

3 Neural Architecture as an Adaptive Surface

We map the training of a neural network to the CDFD life number equation:

$$\Psi_s = \left(\frac{\Phi}{C} \right) \cdot S \cdot M_s \quad (1)$$

In an artificial intelligence system:

- **Flux** (Φ): The gradient updates and the raw throughput of training tokens (data flow).
- **Constraint** (C): The rigid, fixed architecture of the network (e.g., fixed number of layers and attention heads), as well as the physical hardware limits (memory bandwidth and thermal dissipation).
- **Responsiveness** (S): The available parameter space (model width and depth) that can actively adapt to the incoming gradients.
- **Memory** (M_s): The accumulated, frozen weights of the network. As the model trains, it transitions from a high-entropy chaotic state into a low-entropy locked state (overfitting or saturation).

3.1 The Saturation Threshold and Catastrophic Forgetting

During early training, S is high (many free parameters) and M_s is low. The network efficiently absorbs the flux Φ . $\Psi_s \approx 1$.

However, as training progresses, the weights solidify into fixed representations. The systemic memory M_s increases, locking the topology. If new, out-of-distribution data (Φ) is introduced, the fixed architectural constraints (C) cannot flex to accommodate it. The network enters an over-constrained regime ($\Psi_s \ll 1$). To learn the new data, it must overwrite its existing weights,

destroying its previous capabilities—a phenomenon known in machine learning as ”catastrophic forgetting.”

Unlike biological brains, which dynamically grow new synapses and prune old ones (active restructuring of C and S), current LLMs have fixed, static constraints.

4 The Path to General Intelligence

The CDFD framework models the condition that that you cannot achieve infinite adaptability in a system with fixed static constraints. If C is locked, Ψ_s will enter an overload regime as M_s saturates.

True Artificial General Intelligence (AGI) requires an architecture where the constraint C itself is a dynamic variable governed by the differential equation $\frac{\partial C}{\partial t} = \alpha|\Phi| - \beta C$. The network must be able to physically or logically grow new architectural branches (neurogenesis) and prune useless nodes to continually maintain $\Psi_s \approx 1$. Liquid neural networks and sparsely activated mixture-of-experts (MoE) are primitive steps toward this, but an AGI must be a fully self-regulating CDFD surface.

5 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system’s ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

5.1 Current Runtime Diagnostic: Universal Network Cascade

The diagnostic script `Part_E_Synthesis/supplementary/run_partiv_discovery.py` keeps this paper tied to the current CDFD Runtime [4], including RK4 field updates, surface dynamics, structural-memory diffusion, and a finite-value audit. In the May 2026 run, the localized hub-drive stress test ended with hub $\Psi_s = 14.8$, peak network $\Psi_s = 14.8$, 21 nodes above $\Psi_s > 1.5$, and 37 maximum memory-locked nodes. I treat those numbers as a runtime sanity check and as a target for later domain measurement, not as a substitute for field data.

6 Conclusion

Artificial Intelligence is bound by the exact same thermodynamic and topological limits as biological metabolism and galactic formation. By treating deep learning through the lens of Adaptive Flux Limitation, we identify the mathematical ceiling of static neural scaling and outline the required structural dynamics for genuine, open-ended machine cognition.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Release-local runtime diagnostics are available under each Part's `outputs/` folder, with release-wide synthesis artifacts under `Part_E_Synthesis/supplementary/`, `Part_E_Synthesis/outputs/`, and `Part_E_Synthesis/figures/`.

7 Reference Layer

References

- [1] Steve Bico Mujjabi. Cdfd part i: Fundamental physics - constraint-driven field theory and flux dynamics, 2026.
- [2] Steve Bico Mujjabi. Cdfd part ii: Origins of life and tri-regime bioenergetics - constraint-driven flux dynamics, 2026.
- [3] Steve Bico Mujjabi. Cdfd part iii: Adaptive flux limitation biology and medicine - constraint-driven flux dynamics, 2026.
- [4] Steve Bico Mujjabi. Cdfd runtime: Constraint-driven flux dynamics and cdfl execution engine, 2026.